HireSync

University Placement Management System

A Java-based console application bridging students and recruitment companies through automated eligibility verification and application tracking.

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System Overview

Automated Verification

Real-time eligibility checks based on CGPA and backlog criteria

Application Tracking

Complete lifecycle management from application to offer

Business Rules

Enforces fair placement practices and prevents data inconsistency

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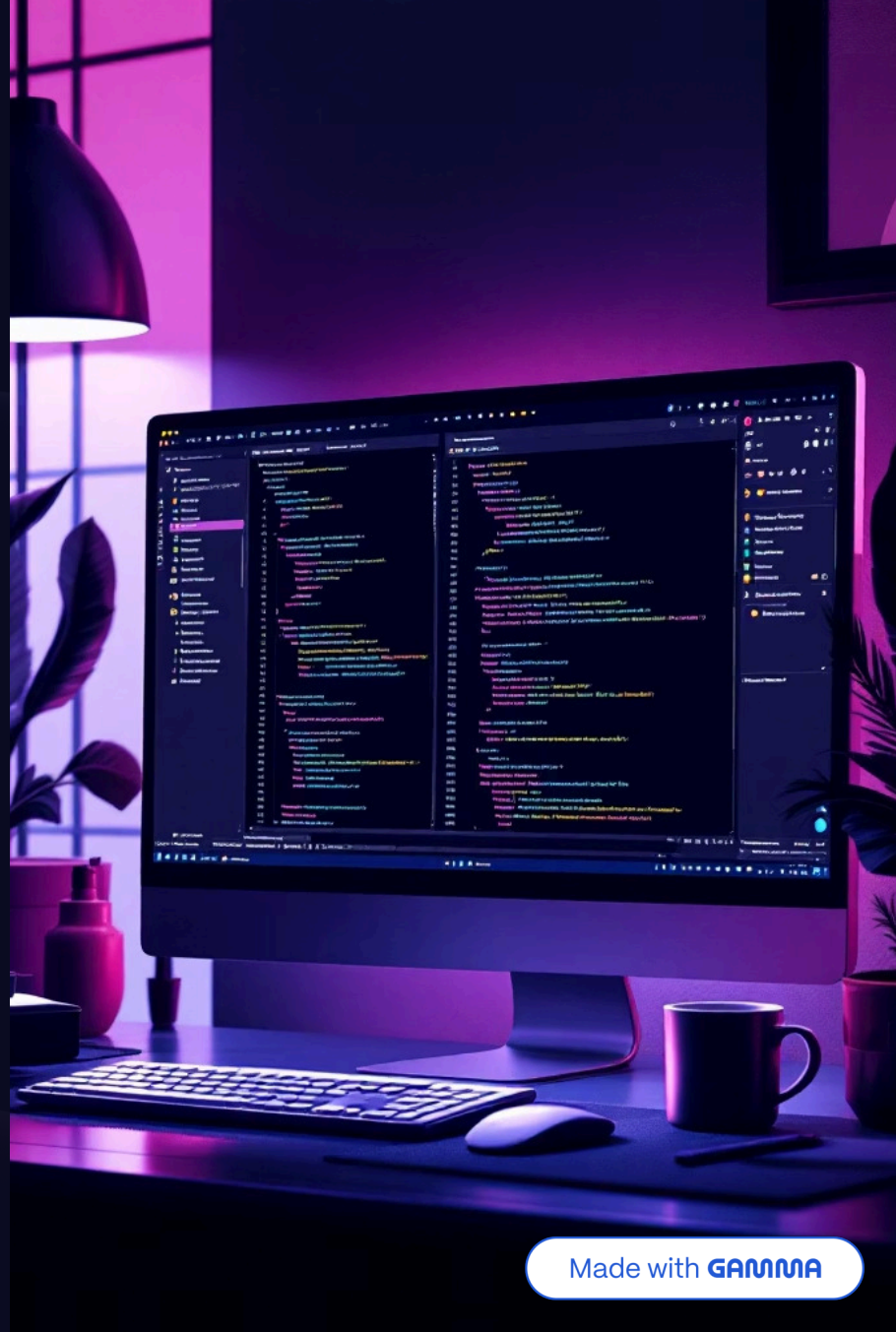
Technology Stack

Core Technology

Built with Core Java (JDK 17) using modular package structure for separation of concerns.

Key Features

- In-memory data storage
- Console-based user interface
- Custom exception handling
- Dynamic filtering capabilities



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Design Patterns Implemented

1

Singleton Pattern

Ensures a single PlacementCoordinator manages the entire ecosystem, preventing data inconsistency across the system.

2

Filter (Criteria) Pattern

Decouples eligibility logic (CGPA, backlogs) from student and job classes, allowing for dynamic filtering.

3

State Pattern

Manages application lifecycle (Applied→Interview→Offer) and prevents illegal transitions between stages.

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Core Modules

Student Profiles

Comprehensive academic metrics including CGPA and active backlog counts for eligibility determination.

Company Management

Organizations can list multiple job profiles, each defining its own selection thresholds.

Eligibility Engine

Real-time validation using Filter Design Pattern to match students with suitable opportunities.

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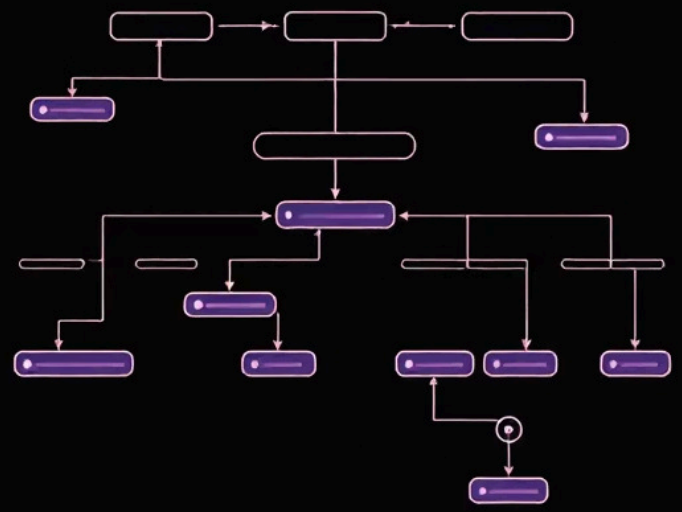

Eligibility Logic

The system performs real-time validation using the Filter Design Pattern to determine student eligibility for each job profile.

1

CGPA Check

Student CGPA > Job MinCGPA

2

Backlog Check

Student Backlogs ≤ Job MaxBacklogs

3

Result

Eligible or Ineligible status

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Application Lifecycle

The State Pattern manages transitions through specific stages, maintaining hiring process integrity and preventing illegal state changes.

Applied

Student submits application to job profile

1

2

Interview

Company reviews and schedules interview

Offer

Final job offer extended to student

3

Key Enforcement: Students cannot receive an offer without first passing the interview stage.

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Business Rule Enforcement

Exclusivity Rule

Students restricted to one active offer to prevent job hoarding and ensure maximum placement across the batch.

State Integrity

Attempts to skip recruitment stages trigger custom exceptions like IneligibleStudentException and DuplicateApplicationException.

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Implementation Details

Package Structure

- `hiresync.models` - Entity POJOs
- `hiresync.state` - State classes
- `hiresync.filters` - Eligibility logic
- `hiresync.exceptions` - Custom errors

Compilation & Execution

```
javac -d bin -sourcepath src
src/hiresync/ui/ConsoleApp.java
java -cp bin hiresync.ui.ConsoleApp
```

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Future Enhancements

Database Integration

Migrate from in-memory storage to MySQL or PostgreSQL for persistent data management.

GUI Interface

Transition from console-based UI to JavaFX or web-based interface for improved user experience.

Resume Parsing

Implement PDF analysis for skill-based filtering and enhanced matching capabilities.

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